

量子力学Ⅱ 第7-9回 確認問題 解答例

3.

(1)

$$i\hbar \frac{\partial}{\partial t} \psi(t) = \hat{H} \psi(t)$$

$$i\hbar \frac{\partial}{\partial t} \begin{pmatrix} \psi_+(t) \\ \psi_-(t) \end{pmatrix} = \mu_B B \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} \psi_+(t) \\ \psi_-(t) \end{pmatrix}$$

(2)

(1) より

$$\frac{d\psi_+(t)}{dt} = -i \frac{\mu_B B}{\hbar} \psi_+(t), \quad \frac{d\psi_-(t)}{dt} = i \frac{\mu_B B}{\hbar} \psi_-(t)$$

これを解くと

$$\psi_+(t) = \psi_+(0) e^{-i \frac{\mu_B B}{\hbar} t}, \quad \psi_-(t) = \psi_-(0) e^{i \frac{\mu_B B}{\hbar} t}$$

$\mathbf{n} = (\sin \theta \cos \varphi, \sin \theta \sin \varphi, \cos \theta)$ とおくと

$$\psi(0) = \begin{pmatrix} \psi_+(0) \\ \psi_-(0) \end{pmatrix} = \begin{pmatrix} \cos \frac{\theta}{2} e^{-\frac{1}{2} i \varphi} \\ \sin \frac{\theta}{2} e^{\frac{1}{2} i \varphi} \end{pmatrix}$$

と選べるから, $\omega = \frac{2\mu_B B}{\hbar}$ として

$$\psi_+(t) = \cos \frac{\theta}{2} e^{-\frac{1}{2} i(\varphi + \omega t)}, \quad \psi_-(t) = \sin \frac{\theta}{2} e^{\frac{1}{2} i(\varphi + \omega t)}$$

すなわち

$$\psi(t) = \begin{pmatrix} \cos \frac{\theta}{2} e^{-\frac{1}{2} i(\varphi + \omega t)} \\ \sin \frac{\theta}{2} e^{\frac{1}{2} i(\varphi + \omega t)} \end{pmatrix}$$

(3)

$$\begin{aligned} \langle \hat{s}_1 \rangle &= (\psi(t), \hat{s}_1 \psi(t)) \\ &= \frac{\hbar}{2} \begin{pmatrix} \cos \frac{\theta}{2} e^{-\frac{1}{2} i(\varphi + \omega t)} \\ \sin \frac{\theta}{2} e^{\frac{1}{2} i(\varphi + \omega t)} \end{pmatrix}^\dagger \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \cos \frac{\theta}{2} e^{-\frac{1}{2} i(\varphi + \omega t)} \\ \sin \frac{\theta}{2} e^{\frac{1}{2} i(\varphi + \omega t)} \end{pmatrix} \\ &= \frac{\hbar}{2} \sin \theta \cos(\varphi + \omega t) \end{aligned}$$

$$\begin{aligned}
\langle \hat{s}_2 \rangle &= (\psi(t), \hat{s}_2 \psi(t)) \\
&= \frac{\hbar}{2} \begin{pmatrix} \cos \frac{\theta}{2} e^{-\frac{1}{2}i(\varphi+\omega t)} \\ \sin \frac{\theta}{2} e^{\frac{1}{2}i(\varphi+\omega t)} \end{pmatrix}^\dagger \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} \cos \frac{\theta}{2} e^{-\frac{1}{2}i(\varphi+\omega t)} \\ \sin \frac{\theta}{2} e^{\frac{1}{2}i(\varphi+\omega t)} \end{pmatrix} \\
&= \frac{\hbar}{2} \sin \theta \sin (\varphi + \omega t)
\end{aligned}$$

$$\begin{aligned}
\langle \hat{s}_3 \rangle &= (\psi(t), \hat{s}_3 \psi(t)) \\
&= \frac{\hbar}{2} \begin{pmatrix} \cos \frac{\theta}{2} e^{-\frac{1}{2}i(\varphi+\omega t)} \\ \sin \frac{\theta}{2} e^{\frac{1}{2}i(\varphi+\omega t)} \end{pmatrix}^\dagger \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} \cos \frac{\theta}{2} e^{-\frac{1}{2}i(\varphi+\omega t)} \\ \sin \frac{\theta}{2} e^{\frac{1}{2}i(\varphi+\omega t)} \end{pmatrix} \\
&= \frac{\hbar}{2} \cos \theta
\end{aligned}$$

したがって、 $\langle \hat{\mathbf{s}} \rangle$ は 3 軸 (z 軸) のまわりを角速度 ω で歳差運動する。

1. (1)

$$\begin{aligned}\sigma_1\sigma_2 + \sigma_2\sigma_1 &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} + \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \\ &= \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} + \begin{pmatrix} -i & 0 \\ 0 & i \end{pmatrix} = 0 \text{ (零行列)}\end{aligned}$$

$$\begin{aligned}\sigma_1\sigma_2 - \sigma_2\sigma_1 &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} - \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \\ &= \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} - \begin{pmatrix} -i & 0 \\ 0 & i \end{pmatrix} \\ &= 2i \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = 2i\sigma_3 = 2i\epsilon_{12k}\sigma_k\end{aligned}$$

$$\begin{aligned}\sigma_2\sigma_3 + \sigma_3\sigma_2 &= \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \\ &= \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix} + \begin{pmatrix} 0 & -i \\ -i & 0 \end{pmatrix} = 0 \text{ (零行列)}\end{aligned}$$

$$\begin{aligned}\sigma_2\sigma_3 - \sigma_3\sigma_2 &= \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} - \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \\ &= \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix} - \begin{pmatrix} 0 & -i \\ -i & 0 \end{pmatrix} \\ &= 2i \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = 2i\sigma_1 = 2i\epsilon_{23k}\sigma_k\end{aligned}$$

$$\begin{aligned}\sigma_3\sigma_1 + \sigma_1\sigma_3 &= \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \\ &= \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} + \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = 0 \text{ (零行列)}\end{aligned}$$

$$\begin{aligned}\sigma_3\sigma_1 - \sigma_1\sigma_3 &= \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \\ &= \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} - \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \\ &= 2i \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = 2i\sigma_2 = 2i\epsilon_{31k}\sigma_k\end{aligned}$$

(2) (1)より

$$[\hat{s}_i, \hat{s}_j] = \left[\frac{\hbar}{2} \sigma_i, \frac{\hbar}{2} \sigma_j \right] = \frac{\hbar^2}{4} 2i\epsilon_{ijk} \sigma_k = i\hbar \epsilon_{ijk} \hat{s}_k$$

(3)

$$\mathbf{n} \cdot \boldsymbol{\sigma} = n_1 \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + n_2 \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} + n_3 \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} n_3 & n_1 - in_2 \\ n_1 + in_2 & -n_3 \end{pmatrix}$$

の固有値を求める.

$$n_1 \pm in_2 = \sqrt{n_1^2 + n_2^2} e^{\pm i\phi} = n' e^{\pm i\phi}$$

とおき, 固有値を λ , 固有ベクトルを

$$u = \begin{pmatrix} a \\ b \end{pmatrix}$$

とおくと

$$\mathbf{n} \cdot \boldsymbol{\sigma} u = \lambda u$$

$$(\mathbf{n} \cdot \boldsymbol{\sigma} - \lambda I) u = 0$$

$u \neq 0$ より $(\mathbf{n} \cdot \boldsymbol{\sigma} - \lambda I)$ は逆行列をもたない. よって

$$|\mathbf{n} \cdot \boldsymbol{\sigma} - \lambda I| = \begin{vmatrix} n_3 - \lambda & n' e^{-i\phi} \\ n' e^{i\phi} & -n_3 - \lambda \end{vmatrix} = 0$$

$$-(n_3 - \lambda)(n_3 + \lambda) - n'^2 = 0$$

$$\lambda^2 = n'^2 + n_3^2 = n_1^2 + n_2^2 + n_3^2 = 1$$

となり, $\lambda = \pm 1$. すなわち, $\mathbf{n} \cdot \hat{\mathbf{s}}$ の固有値は $\pm \frac{\hbar}{2}$

$\lambda = 1$ に対する固有ベクトル u_1 は

$$\begin{pmatrix} n_3 - 1 & n' e^{-i\phi} \\ n' e^{i\phi} & -n_3 - 1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$(n_3 - 1)a + n' e^{-i\phi} b = 0$$

$$a:b = n' e^{-i\phi} : (1 - n_3) = n_1 - in_2 : 1 - n_3$$

$$|n_1 - in_2|^2 + (1 - n_3)^2 = n_1^2 + n_2^2 + 1 - 2n_3 + n_3^2 = 2(1 - n_3)$$

より, 規格化された固有ベクトルは

$$u_1 = \frac{1}{\sqrt{2(1 - n_3)}} \begin{pmatrix} n_1 - in_2 \\ 1 - n_3 \end{pmatrix}$$

$\lambda = -1$ に対する固有ベクトル u_{-1} は

$$\begin{pmatrix} n_3 + 1 & n' e^{-i\phi} \\ n' e^{i\phi} & -n_3 + 1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$(n_3 + 1)a + n' e^{-i\phi} b = 0$$

$$a:b = n' e^{-i\phi} : -(1 + n_3) = n_1 - in_2 : -(1 + n_3)$$

$$|n_1 - in_2|^2 + (1 + n_3)^2 = n_1^2 + n_2^2 + 1 + 2n_3 + n_3^2 = 2(1 + n_3)$$

より、規格化された固有ベクトルは

$$u_{-1} = \frac{1}{\sqrt{2(1 + n_3)}} \begin{pmatrix} -(n_1 - in_2) \\ 1 + n_3 \end{pmatrix}$$

$\mathbf{n} = (n_1, n_2, n_3)$ の成分を極座標で

$$n_1 = \sin \theta \cos \phi, \quad n_2 = \sin \theta \sin \phi, \quad n_3 = \cos \theta$$

と表すと

$$n_1 \pm in_2 = \sin \theta e^{\pm i\phi} = 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2} e^{\pm i\phi}$$

$$1 + n_3 = 1 + \cos \theta = 2 \cos^2 \frac{\theta}{2}, \quad 1 - n_3 = 1 - \cos \theta = 2 \sin^2 \frac{\theta}{2}$$

より

$$u_1 = \begin{pmatrix} \cos \frac{\theta}{2} e^{-i\phi} \\ \sin \frac{\theta}{2} \end{pmatrix}, \quad u_{-1} = \begin{pmatrix} -\sin \frac{\theta}{2} e^{-i\phi} \\ \cos \frac{\theta}{2} \end{pmatrix}$$

と表すこともできる.

(補足)

$\mathbf{n}_0 = (0, 0, 1)$ を x_2 軸回りに θ 回転し、続けて x_3 軸回りに ϕ 回転すると $\mathbf{n}$ となる.

$$\mathbf{n}_0 \cdot \hat{\mathbf{s}} = \frac{\hbar}{2} \sigma_3$$

より、 $\mathbf{n}_0 \cdot \hat{\mathbf{s}}$ の固有値は $\pm \frac{\hbar}{2}$

$\mathbf{n} \cdot \hat{\mathbf{s}}$ は原点のまわりの回転に対しスカラー量だから、その固有値は回転により変化しない.

したがって、 $\mathbf{n} \cdot \hat{\mathbf{s}}$ の固有値は、 $\mathbf{n}_0 \cdot \hat{\mathbf{s}}$ の固有値と同じく $\pm \frac{\hbar}{2}$

一方、固有値 $\pm \frac{\hbar}{2}$ に対する $\mathbf{n}_0 \cdot \hat{\mathbf{s}}$ の固有ベクトルは、それぞれ

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

であり、これらは回転により

$$e^{-i\frac{\hat{s}_3}{\hbar}\phi} e^{-i\frac{\hat{s}_2}{\hbar}\theta} \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad e^{-i\frac{\hat{s}_3}{\hbar}\phi} e^{-i\frac{\hat{s}_2}{\hbar}\theta} \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

に変化する. ここで

$$e^{-i\frac{\hat{s}_3}{\hbar}\phi} e^{-i\frac{\hat{s}_2}{\hbar}\theta} = e^{-i\frac{\sigma_3}{2}\phi} e^{-i\frac{\sigma_2}{2}\theta} = \begin{pmatrix} e^{-i\frac{\phi}{2}} & 0 \\ 0 & e^{i\frac{\phi}{2}} \end{pmatrix} \begin{pmatrix} \cos \frac{\theta}{2} & -\sin \frac{\theta}{2} \\ \sin \frac{\theta}{2} & \cos \frac{\theta}{2} \end{pmatrix}$$

であるから，回転後の固有ベクトルはそれぞれ

$$\begin{pmatrix} e^{-\frac{i\phi}{2}} & 0 \\ 0 & e^{\frac{i\phi}{2}} \end{pmatrix} \begin{pmatrix} \cos \frac{\theta}{2} & -\sin \frac{\theta}{2} \\ \sin \frac{\theta}{2} & \cos \frac{\theta}{2} \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} \cos \frac{\theta}{2} e^{-\frac{i\phi}{2}} \\ \sin \frac{\theta}{2} e^{\frac{i\phi}{2}} \end{pmatrix} = e^{\frac{i\phi}{2}} \begin{pmatrix} \cos \frac{\theta}{2} e^{-i\phi} \\ \sin \frac{\theta}{2} \end{pmatrix}$$
$$\begin{pmatrix} e^{-\frac{i\phi}{2}} & 0 \\ 0 & e^{\frac{i\phi}{2}} \end{pmatrix} \begin{pmatrix} \cos \frac{\theta}{2} & -\sin \frac{\theta}{2} \\ \sin \frac{\theta}{2} & \cos \frac{\theta}{2} \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -\sin \frac{\theta}{2} e^{-\frac{i\phi}{2}} \\ \cos \frac{\theta}{2} e^{\frac{i\phi}{2}} \end{pmatrix} = e^{\frac{i\phi}{2}} \begin{pmatrix} -\sin \frac{\theta}{2} e^{-i\phi} \\ \cos \frac{\theta}{2} \end{pmatrix}$$

となり，直接計算により求めた固有ベクトルと一致する．

2.

$$\hat{J}_z|j, m\rangle = m\hbar|j, m\rangle$$

$$\hat{J}_+|j, m\rangle = \sqrt{j(j+1) - m(m+1)}\hbar|j, m+1\rangle, \quad \hat{J}_-|j, m\rangle = \sqrt{j(j+1) - m(m-1)}\hbar|j, m-1\rangle$$

などを用いる.

(1)

$|l=1, m=1\rangle \left| s=\frac{1}{2}, m_s=\frac{1}{2} \right\rangle$ に $\hat{J}_z = \hat{L}_z + \hat{S}_z$ を作用させると

$$\begin{aligned} \hat{J}_z|l=1, m=1\rangle \left| s=\frac{1}{2}, m_s=\frac{1}{2} \right\rangle &= (\hat{L}_z + \hat{S}_z)|l=1, m=1\rangle \left| s=\frac{1}{2}, m_s=\frac{1}{2} \right\rangle \\ &= (\hat{L}_z|l=1, m=1\rangle) \left| s=\frac{1}{2}, m_s=\frac{1}{2} \right\rangle + |l=1, m=1\rangle (\hat{S}_z \left| s=\frac{1}{2}, m_s=\frac{1}{2} \right\rangle) \\ &= (\hbar|l=1, m=1\rangle) \left| s=\frac{1}{2}, m_s=\frac{1}{2} \right\rangle + |l=1, m=1\rangle \left(\frac{\hbar}{2} \left| s=\frac{1}{2}, m_s=\frac{1}{2} \right\rangle \right) \\ &= \frac{3}{2}\hbar|l=1, m=1\rangle \left| s=\frac{1}{2}, m_s=\frac{1}{2} \right\rangle \quad \text{①} \end{aligned}$$

$|l=1, m=1\rangle \left| s=\frac{1}{2}, m_s=\frac{1}{2} \right\rangle$ に $\hat{J}_+ = \hat{L}_+ + \hat{S}_+$ を作用させると

$$\begin{aligned} \hat{J}_+|l=1, m=1\rangle \left| s=\frac{1}{2}, m_s=\frac{1}{2} \right\rangle &= (\hat{L}_+ + \hat{S}_+)|l=1, m=1\rangle \left| s=\frac{1}{2}, m_s=\frac{1}{2} \right\rangle \\ &= (\hat{L}_+|l=1, m=1\rangle) \left| s=\frac{1}{2}, m_s=\frac{1}{2} \right\rangle + |l=1, m=1\rangle (\hat{S}_+ \left| s=\frac{1}{2}, m_s=\frac{1}{2} \right\rangle) \\ &= (0) \left| s=\frac{1}{2}, m_s=\frac{1}{2} \right\rangle + |l=1, m=1\rangle (0) = 0 \end{aligned}$$

よって

$$\begin{aligned} \hat{J}^2|l=1, m=1\rangle \left| s=\frac{1}{2}, m_s=\frac{1}{2} \right\rangle &= (\hat{J}_- \hat{J}_+ + \hat{J}_z^2 + \hbar \hat{J}_z)|l=1, m=1\rangle \left| s=\frac{1}{2}, m_s=\frac{1}{2} \right\rangle \\ &= \left\{ \left(\frac{3}{2}\hbar \right)^2 + \hbar \cdot \frac{3}{2}\hbar \right\} |l=1, m=1\rangle \left| s=\frac{1}{2}, m_s=\frac{1}{2} \right\rangle \\ &= \frac{3}{2} \left(\frac{3}{2} + 1 \right) \hbar^2 |l=1, m=1\rangle \left| s=\frac{1}{2}, m_s=\frac{1}{2} \right\rangle \quad \text{②} \end{aligned}$$

①, ②より

$$\left| j=\frac{3}{2}, m_j=\frac{3}{2} \right\rangle = |l=1, m=1\rangle \left| s=\frac{1}{2}, m_s=\frac{1}{2} \right\rangle$$

(2)

$|j = \frac{3}{2}, m_j = \frac{3}{2}\rangle = |l = 1, m = 1\rangle |s = \frac{1}{2}, m_s = \frac{1}{2}\rangle$ に $\hat{j}_- = \hat{l}_- + \hat{s}_-$ を作用させると

$$\begin{aligned} \hat{j}_- |j = \frac{3}{2}, m_j = \frac{3}{2}\rangle &= (\hat{l}_- + \hat{s}_-) |l = 1, m = 1\rangle |s = \frac{1}{2}, m_s = \frac{1}{2}\rangle \\ &= (\hat{l}_- |l = 1, m = 1\rangle) |s = \frac{1}{2}, m_s = \frac{1}{2}\rangle + |l = 1, m = 1\rangle (\hat{s}_- |s = \frac{1}{2}, m_s = \frac{1}{2}\rangle) \\ &= \left(\sqrt{\frac{3}{2} \left(\frac{3}{2} + 1 \right) - \frac{3}{2} \left(\frac{3}{2} - 1 \right)} \hbar |j = \frac{3}{2}, m_j = \frac{1}{2}\rangle \right. \\ &\quad \left. + |l = 1, m = 1\rangle \left(\sqrt{\frac{1}{2} \left(\frac{1}{2} + 1 \right) - \frac{1}{2} \left(\frac{1}{2} - 1 \right)} \hbar |s = \frac{1}{2}, m_s = -\frac{1}{2}\rangle \right) \right) \end{aligned}$$

よって

$$|j = \frac{3}{2}, m_j = \frac{1}{2}\rangle = \sqrt{\frac{2}{3}} |l = 1, m = 0\rangle |s = \frac{1}{2}, m_s = \frac{1}{2}\rangle + \sqrt{\frac{1}{3}} |l = 1, m = 1\rangle |s = \frac{1}{2}, m_s = -\frac{1}{2}\rangle$$

この式にさらに $\hat{j}_- = \hat{l}_- + \hat{s}_-$ を作用させると

$$\begin{aligned} \hat{j}_- |j = \frac{3}{2}, m_j = \frac{1}{2}\rangle &= \sqrt{\frac{2}{3}} (\hat{l}_- + \hat{s}_-) |l = 1, m = 0\rangle |s = \frac{1}{2}, m_s = \frac{1}{2}\rangle \\ &\quad + \sqrt{\frac{1}{3}} (\hat{l}_- + \hat{s}_-) |l = 1, m = 1\rangle |s = \frac{1}{2}, m_s = -\frac{1}{2}\rangle \\ &= \sqrt{\frac{2}{3}} (\hat{l}_- |l = 1, m = 0\rangle) |s = \frac{1}{2}, m_s = \frac{1}{2}\rangle + \sqrt{\frac{2}{3}} |l = 1, m = 0\rangle (\hat{s}_- |s = \frac{1}{2}, m_s = \frac{1}{2}\rangle) \\ &\quad + \sqrt{\frac{1}{3}} (\hat{l}_- |l = 1, m = 1\rangle) |s = \frac{1}{2}, m_s = -\frac{1}{2}\rangle + \sqrt{\frac{1}{3}} |l = 1, m = 1\rangle (\hat{s}_- |s = \frac{1}{2}, m_s = -\frac{1}{2}\rangle) \\ &= \sqrt{\frac{2}{3}} \left(\sqrt{1(1+1) - 0(0-1)} \hbar |l = 1, m = -1\rangle \right) |s = \frac{1}{2}, m_s = \frac{1}{2}\rangle \\ &\quad + \sqrt{\frac{2}{3}} |l = 1, m = 0\rangle \left(\sqrt{\frac{1}{2} \left(\frac{1}{2} + 1 \right) - \frac{1}{2} \left(\frac{1}{2} - 1 \right)} \hbar |s = \frac{1}{2}, m_s = -\frac{1}{2}\rangle \right) \end{aligned}$$

$$\begin{aligned}
& + \sqrt{\frac{1}{3}} (\sqrt{1(1+1) - 1(1-1)} \hbar |l=1, m=0\rangle) \left| s=\frac{1}{2}, m_s=-\frac{1}{2} \right\rangle \\
& + \sqrt{\frac{1}{3}} |l=1, m=1\rangle (0)
\end{aligned}$$

よって

$$\left| j=\frac{3}{2}, m_j=-\frac{1}{2} \right\rangle = \sqrt{\frac{1}{3}} |l=1, m=-1\rangle \left| s=\frac{1}{2}, m_s=\frac{1}{2} \right\rangle + \sqrt{\frac{2}{3}} |l=1, m=0\rangle \left| s=\frac{1}{2}, m_s=-\frac{1}{2} \right\rangle$$

この式にさらに $\hat{j}_- = \hat{l}_- + \hat{s}_-$ を作用させると

$$\begin{aligned}
\hat{j}_- \left| j=\frac{3}{2}, m_j=-\frac{1}{2} \right\rangle &= \sqrt{\frac{1}{3}} (\hat{l}_- + \hat{s}_-) |l=1, m=-1\rangle \left| s=\frac{1}{2}, m_s=\frac{1}{2} \right\rangle \\
&+ \sqrt{\frac{2}{3}} (\hat{l}_- + \hat{s}_-) |l=1, m=0\rangle \left| s=\frac{1}{2}, m_s=-\frac{1}{2} \right\rangle \\
&= \sqrt{\frac{1}{3}} (\hat{l}_- |l=1, m=-1\rangle) \left| s=\frac{1}{2}, m_s=\frac{1}{2} \right\rangle + \sqrt{\frac{1}{3}} |l=1, m=-1\rangle (\hat{s}_- \left| s=\frac{1}{2}, m_s=\frac{1}{2} \right\rangle) \\
&+ \sqrt{\frac{2}{3}} (\hat{l}_- |l=1, m=0\rangle) \left| s=\frac{1}{2}, m_s=-\frac{1}{2} \right\rangle + \sqrt{\frac{2}{3}} |l=1, m=0\rangle (\hat{s}_- \left| s=\frac{1}{2}, m_s=-\frac{1}{2} \right\rangle)
\end{aligned}$$

$$\begin{aligned}
& \sqrt{\frac{3}{2} \left(\frac{3}{2} + 1 \right) - \left(-\frac{1}{2} \right) \left(-\frac{1}{2} - 1 \right)} \hbar \left| j=\frac{3}{2}, m_j=-\frac{3}{2} \right\rangle \\
&= \sqrt{\frac{1}{3}} (0) \left| s=\frac{1}{2}, m_s=\frac{1}{2} \right\rangle \\
&+ \sqrt{\frac{1}{3}} |l=1, m=-1\rangle \left(\sqrt{\frac{1}{2} \left(\frac{1}{2} + 1 \right) - \frac{1}{2} \left(\frac{1}{2} - 1 \right)} \hbar \left| s=\frac{1}{2}, m_s=-\frac{1}{2} \right\rangle \right) \\
&+ \sqrt{\frac{2}{3}} (\sqrt{1(1+1) - 0(0-1)} \hbar |l=1, m=-1\rangle) \left| s=\frac{1}{2}, m_s=-\frac{1}{2} \right\rangle \\
&+ \sqrt{\frac{2}{3}} |l=1, m=0\rangle (0)
\end{aligned}$$

よって

$$\left| j=\frac{3}{2}, m_j=-\frac{3}{2} \right\rangle = |l=1, m=-1\rangle \left| s=\frac{1}{2}, m_s=-\frac{1}{2} \right\rangle$$

(3)

$$|\phi\rangle = \sqrt{\frac{1}{3}} |l=1, m=0\rangle \left| s=\frac{1}{2}, m_s=\frac{1}{2} \right\rangle - \sqrt{\frac{2}{3}} |l=1, m=1\rangle \left| s=\frac{1}{2}, m_s=-\frac{1}{2} \right\rangle$$

に $\hat{j}_z = \hat{l}_z + \hat{s}_z$ を作用させると

$$\begin{aligned}
\hat{j}_z|\phi\rangle &= \sqrt{\frac{1}{3}}(\hat{l}_z + \hat{s}_z)|l=1, m=0\rangle \left|s=\frac{1}{2}, m_s=\frac{1}{2}\right\rangle - \sqrt{\frac{2}{3}}(\hat{l}_z + \hat{s}_z)|l=1, m=1\rangle \left|s=\frac{1}{2}, m_s=-\frac{1}{2}\right\rangle \\
&= \sqrt{\frac{1}{3}}(\hat{l}_z|l=1, m=0\rangle) \left|s=\frac{1}{2}, m_s=\frac{1}{2}\right\rangle + \sqrt{\frac{1}{3}}|l=1, m=0\rangle(\hat{s}_z \left|s=\frac{1}{2}, m_s=\frac{1}{2}\right\rangle) \\
&\quad - \sqrt{\frac{2}{3}}(\hat{l}_z|l=1, m=1\rangle) \left|s=\frac{1}{2}, m_s=-\frac{1}{2}\right\rangle - \sqrt{\frac{2}{3}}|l=1, m=1\rangle(\hat{s}_z \left|s=\frac{1}{2}, m_s=-\frac{1}{2}\right\rangle) \\
&= \sqrt{\frac{1}{3}}(0) \left|s=\frac{1}{2}, m_s=\frac{1}{2}\right\rangle + \sqrt{\frac{1}{3}}|l=1, m=0\rangle\left(\frac{\hbar}{2} \left|s=\frac{1}{2}, m_s=\frac{1}{2}\right\rangle\right) \\
&\quad - \sqrt{\frac{2}{3}}(\hbar|l=1, m=1\rangle) \left|s=\frac{1}{2}, m_s=-\frac{1}{2}\right\rangle - \sqrt{\frac{2}{3}}|l=1, m=1\rangle\left(-\frac{\hbar}{2} \left|s=\frac{1}{2}, m_s=-\frac{1}{2}\right\rangle\right) \\
&= \frac{\hbar}{2} \left(\sqrt{\frac{1}{3}}|l=1, m=0\rangle \left|s=\frac{1}{2}, m_s=\frac{1}{2}\right\rangle - \sqrt{\frac{2}{3}}|l=1, m=1\rangle \left|s=\frac{1}{2}, m_s=-\frac{1}{2}\right\rangle \right) \\
&= \frac{\hbar}{2}|\phi\rangle \quad \textcircled{1}
\end{aligned}$$

$$\begin{aligned}
\hat{j}_+|\phi\rangle &= \sqrt{\frac{1}{3}}(\hat{l}_+ + \hat{s}_+)|l=1, m=0\rangle \left|s=\frac{1}{2}, m_s=\frac{1}{2}\right\rangle \\
&\quad - \sqrt{\frac{2}{3}}(\hat{l}_+ + \hat{s}_+)|l=1, m=1\rangle \left|s=\frac{1}{2}, m_s=-\frac{1}{2}\right\rangle \\
&= \sqrt{\frac{1}{3}}(\hat{l}_+|l=1, m=0\rangle) \left|s=\frac{1}{2}, m_s=\frac{1}{2}\right\rangle + \sqrt{\frac{1}{3}}|l=1, m=0\rangle(\hat{s}_+ \left|s=\frac{1}{2}, m_s=\frac{1}{2}\right\rangle) \\
&\quad - \sqrt{\frac{2}{3}}(\hat{l}_+|l=1, m=1\rangle) \left|s=\frac{1}{2}, m_s=-\frac{1}{2}\right\rangle - \sqrt{\frac{2}{3}}|l=1, m=1\rangle(\hat{s}_+ \left|s=\frac{1}{2}, m_s=-\frac{1}{2}\right\rangle) \\
&= \sqrt{\frac{1}{3}}(\sqrt{1(1+1)-0(0+1)}\hbar|l=1, m=1\rangle) \left|s=\frac{1}{2}, m_s=\frac{1}{2}\right\rangle \\
&\quad + \sqrt{\frac{1}{3}}|l=1, m=0\rangle(0) \\
&\quad - \sqrt{\frac{2}{3}}(0) \left|s=\frac{1}{2}, m_s=-\frac{1}{2}\right\rangle \\
&\quad - \sqrt{\frac{2}{3}}|l=1, m=1\rangle \left(\sqrt{\frac{1}{2}\left(\frac{1}{2}+1\right) - \left(-\frac{1}{2}\right)\left(-\frac{1}{2}+1\right)}\hbar \left|s=\frac{1}{2}, m_s=\frac{1}{2}\right\rangle \right) \\
\hat{j}_+|\phi\rangle &= 0
\end{aligned}$$

したがって

$$\hat{j}^2|\phi\rangle = (\hat{j}_- \hat{j}_+ + \hat{j}_z^2 + \hbar \hat{j}_z)|\phi\rangle = \left\{ \left(\frac{\hbar}{2}\right)^2 + \hbar \cdot \frac{\hbar}{2} \right\} |\phi\rangle = \frac{1}{2} \left(\frac{1}{2} + 1\right) \hbar^2 |\phi\rangle \quad (2)$$

①, ②より

$$|\phi\rangle = \left| j = \frac{1}{2}, m_j = \frac{1}{2} \right\rangle = \sqrt{\frac{1}{3}} |l = 1, m = 0\rangle \left| s = \frac{1}{2}, m_s = \frac{1}{2} \right\rangle - \sqrt{\frac{2}{3}} |l = 1, m = 1\rangle \left| s = \frac{1}{2}, m_s = -\frac{1}{2} \right\rangle$$

(4)

$$\left| j = \frac{1}{2}, m_j = \frac{1}{2} \right\rangle = \sqrt{\frac{1}{3}} |l = 1, m = 0\rangle \left| s = \frac{1}{2}, m_s = \frac{1}{2} \right\rangle - \sqrt{\frac{2}{3}} |l = 1, m = 1\rangle \left| s = \frac{1}{2}, m_s = -\frac{1}{2} \right\rangle$$

に $\hat{j}_- = \hat{l}_- + \hat{s}_-$ を作用させると

$$\begin{aligned} \hat{j}_- \left| j = \frac{1}{2}, m_j = \frac{1}{2} \right\rangle &= \sqrt{\frac{1}{3}} (\hat{l}_- + \hat{s}_-) |l = 1, m = 0\rangle \left| s = \frac{1}{2}, m_s = \frac{1}{2} \right\rangle \\ &\quad - \sqrt{\frac{2}{3}} (\hat{l}_- + \hat{s}_-) |l = 1, m = 1\rangle \left| s = \frac{1}{2}, m_s = -\frac{1}{2} \right\rangle \\ &= \sqrt{\frac{1}{3}} (\hat{l}_- |l = 1, m = 0\rangle) \left| s = \frac{1}{2}, m_s = \frac{1}{2} \right\rangle + \sqrt{\frac{1}{3}} |l = 1, m = 0\rangle (\hat{s}_- \left| s = \frac{1}{2}, m_s = \frac{1}{2} \right\rangle) \\ &\quad - \sqrt{\frac{2}{3}} (\hat{l}_- |l = 1, m = 1\rangle) \left| s = \frac{1}{2}, m_s = -\frac{1}{2} \right\rangle - \sqrt{\frac{2}{3}} |l = 1, m = 1\rangle (\hat{s}_- \left| s = \frac{1}{2}, m_s = -\frac{1}{2} \right\rangle) \\ &\quad \sqrt{\frac{1}{2} \left(\frac{1}{2} + 1\right) - \frac{1}{2} \left(\frac{1}{2} - 1\right)} \hbar \left| j = \frac{1}{2}, m_j = -\frac{1}{2} \right\rangle \\ &= \sqrt{\frac{1}{3}} (\sqrt{1(1+1) - 0(0-1)} \hbar |l = 1, m = -1\rangle) \left| s = \frac{1}{2}, m_s = \frac{1}{2} \right\rangle \\ &\quad + \sqrt{\frac{1}{3}} |l = 1, m = 0\rangle \left(\sqrt{\frac{1}{2} \left(\frac{1}{2} + 1\right) - \frac{1}{2} \left(\frac{1}{2} - 1\right)} \hbar \left| s = \frac{1}{2}, m_s = -\frac{1}{2} \right\rangle \right) \\ &\quad - \sqrt{\frac{2}{3}} (\sqrt{1(1+1) - 1(1-1)} \hbar |l = 1, m = 0\rangle) \left| s = \frac{1}{2}, m_s = -\frac{1}{2} \right\rangle \\ &\quad - \sqrt{\frac{2}{3}} |l = 1, m = 1\rangle (0) \end{aligned}$$

よって

$$\left| j = \frac{1}{2}, m_j = -\frac{1}{2} \right\rangle = \sqrt{\frac{2}{3}} |l = 1, m = -1\rangle \left| s = \frac{1}{2}, m_s = \frac{1}{2} \right\rangle - \sqrt{\frac{1}{3}} |l = 1, m = 0\rangle \left| s = \frac{1}{2}, m_s = -\frac{1}{2} \right\rangle$$